Multi-Reference Voice Cloning for Qwen3-TTS: A Trade-Off Between Speaker Fidelity and Naturalness

Abstract

Zero-shot voice cloning typically uses a single reference utterance to characterise a target speaker. When multiple references are available, it is unclear how best to combine them. We systematically compare two multi-reference strategies for Qwen3-TTS: (1) audio concatenation, which provides multiple audio–text pairs as in-context examples, and (2) embedding averaging, which averages speaker embeddings extracted from each reference. In a controlled experiment of 1,275 runs—5 speakers × 5 targets × 3 seeds × 17 configurations on LibriTTS-R—we find a statistically significant trade-off: audio concatenation improves speaker similarity (SIM $+0.02$, $p < 0.001$, *d up to* 0.62*) while embedding averaging improves naturalness (UTMOS* $+0.06$, $p < 10^{-7}$, *d up to* 0.61*). Neither approach dominates both axes, revealing a Pareto trade-off. Selecting the longest available references consistently outperforms random selection. A pilot perceptual evaluation suggests that automated speaker similarity metrics may not fully capture human voice identity judgments.*

Index Terms: voice cloning, text-to-speech, speaker similarity, few-shot learning, multi-reference

1 Introduction

Zero-shot voice cloning—synthesising speech in a novel speaker's voice from one or a few reference utterances—has advanced rapidly with neural text-to-speech (TTS) models. Systems such as VALL-E [?], Seed-TTS [?], Voicebox [?], and Qwen3-TTS [?] can produce high-quality speech from a single short reference, yet a single utterance may not fully represent a speaker's vocal characteristics.

In practice, multiple reference utterances are often available—from audiobook recordings, podcast episodes, or prior interactions. A natural question arises: how should multiple references be combined to improve voice cloning quality? Two straightforward strategies exist. First, references can be *concatenated* into the model prompt, providing multiple in-context examples from which the model infers the speaker identity. Second, speaker embeddings can be independently *extracted and averaged*, yielding a smoothed representation that is fed to the model. Despite the practical importance of this choice, systematic comparisons under controlled conditions are scarce.

Recent zero-shot TTS systems span diverse architectures: codec language models (VALL-E [?], CLaM-TTS [?]), diffusion-based approaches (NaturalSpeech 3 [?], DiTTo-TTS [?]), and flow-matching models (Voicebox [?], CosyVoice [?]). Multi-speaker adaptation has been explored in YourTTS [?], which fine-tunes on multiple speakers but does not compare inference-time multi-reference strategies. A recent survey [?] highlights that while the field has made rapid progress on zero-shot cloning quality, the question of how to best leverage multiple references at inference time remains largely unexplored.

We address this gap with a large-scale experiment on Qwen3-TTS [?]. Our contributions are:

1. A systematic comparison of two multi-reference voice cloning strategies on a recent neural TTS model across 1,275 controlled runs.
2. The discovery of a statistically significant Pareto trade-off: audio concatenation optimises speaker fidelity while embedding averaging optimises naturalness, with medium-to-large effect sizes.
3. Analysis of scaling behaviour and reference selection strategy, finding that longer references outperform random selection and that performance plateaus at 3 references.

2 Method

2.1 *Multi-reference strategies*

We evaluate three configurations, illustrated in Figure 1.

Single baseline. The standard single-reference approach: one audio–text pair serves as the speaker prompt. This is the default mode for Qwen3-TTS and serves as our control condition.

Audio concatenation (CONCAT). Multiple reference audio–text pairs are concatenated into a single prompt sequence. The model processes all references as in-context examples before generating the target utterance. This leverages the autoregressive model's in-context learning capability, providing richer acoustic evidence of the target speaker.

Embedding averaging (EMBED). A speaker embedding (x-vector) is independently extracted from each reference using the model's internal speaker encoder. The resulting embeddings are averaged element-wise to produce a single speaker vector, which is provided to the model via its `x_vector_only` mode. This aims to capture a more stable speaker representation by averaging out utterance-specific variation.

2.2 *Experimental setup*

Model. We use Qwen3-TTS-12Hz-0.6B-Base [?] in bfloat16 precision on an NVIDIA A10G GPU (24 GB

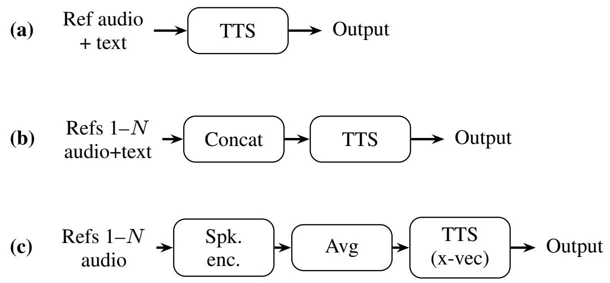

Figure 1: Multi-reference voice cloning strategies. (a) Single baseline: one reference. (b) Audio concatenation: references concatenated as in-context examples. (c) Embedding averaging: embeddings averaged into one x-vector.

VRAM).

Dataset. Evaluation uses LibriTTS-R [?], a restored version of LibriTTS [?], at 24 kHz. We select 5 speakers (IDs: 1188, 1995, 4446, 4992, 5142), allocating 35 reference utterances and 5 held-out target texts per speaker.

Experimental design. We vary: approach $\in$ {single, CONCAT, EMBED}, number of references $n \in \{1, 2, 3, 5\}$, selection strategy $\in$ {random, longest}, and random seed $\in \{42, 123, 456\}$. This yields $5 \times 5 \times 3 \times 17 =$ 1,275 synthesis runs (1 baseline + 8 CONCAT + 8 EMBED configurations).

2.3 Evaluation metrics

Speaker similarity (SIM). Cosine similarity between WavLM-based speaker embeddings [?] of synthesised and ground-truth speech, using the `wavlm-base-plus-sv` model.

Naturalness (UTMOS). Predicted mean opinion score from the UTMOS model [?], an automated proxy for human naturalness judgments.

Word error rate (WER). Whisper-turbo [?] transcription of synthesised audio compared to target text. Because ASR error is non-negligible on ground-truth LibriTTS-R audio, we treat WER primarily as a stability proxy and report calibrated interpretation in Section 3.2.

Statistical testing. All comparisons use paired Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = 0.05$, 48 comparisons). Effect sizes are reported as Cohen's d [?].

3 Results

3.1 Overall comparison

Table 1 presents the full results. CONCAT with the longest-reference strategy achieves the highest speaker similarity (SIM = 0.949 at n=3), an improvement of +0.021 over the single baseline (0.928). EMBED achieves the highest naturalness (UTMOS = 4.483 at n=3, random), an improvement of +0.058 over the baseline (4.425). Neither approach simultaneously improves both metrics: CONCAT maintains baseline-level naturalness while improving fidelity, and EMBED maintains baseline-level similarity while improving naturalness.

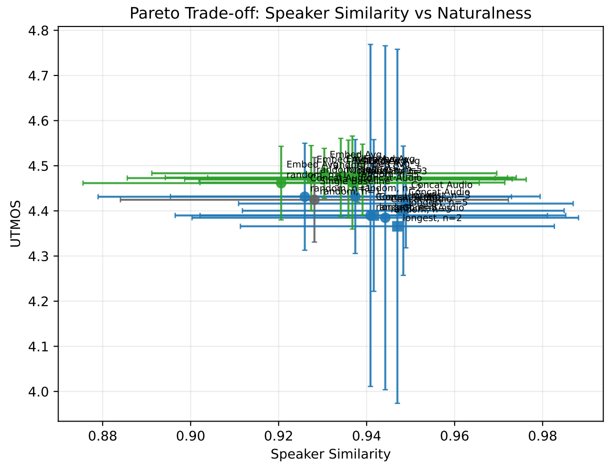

Figure 2: Pareto trade-off between speaker similarity and naturalness across all configurations. CONCAT+longest maximises SIM, while EMBED maximises UTMOS.

3.2 Statistical significance

Table 2 summarises the significant results after Bonferroni correction. For speaker similarity, four CONCAT configurations achieve significance ($p < 0.003$, $d = 0.38$–0.62), three with the longest strategy and one with random. The strongest effects appear with 3 and 5 references ($d = 0.60$ and 0.62).

For naturalness, all eight EMBED configurations achieve significance ($p < 0.004$, $d = 0.32$–0.61). The strongest effect is EMBED with 3 random references ($p < 10^{-7}$, $d = 0.61$).

No configuration produces a statistically significant change in WER after correction. Whisper calibration on the 25 ground-truth target audios yields $\text{WER}_{\text{GT}} = 14.3\%$ median (IQR: 6.5–22.2%), indicating substantial ASR floor error unrelated to synthesis quality. Accordingly, we interpret WER deltas conservatively and focus on catastrophic-failure rates. CONCAT with random selection at $n \geq 3$ shows the clearest instability: failure rate (WER$>$ 50%) of 2.7% at n=3 and 1.3% at n=5, with rare extreme outliers inflating means. By contrast, EMBED has 0% failures under this threshold across all tested configurations.

3.3 Pareto trade-off

Figure 2 visualises the main result directly in SIM–UTMOS space. CONCAT+longest occupies the high-SIM frontier (peak SIM at n=3), while EMBED occupies the high-UTMOS frontier (peak UTMOS at n=3, random). No configuration dominates both axes, confirming the Pareto trade-off and motivating task-specific operating-point selection.

3.4 Reference selection strategy

Figure 3 shows stability through catastrophic-failure rates. The instability risk is concentrated in CONCAT with random selection, particularly at larger reference counts. For

Table 1: Main results across all configurations. Bold indicates best value per metric. † Significant improvement over baseline ($p < 0.05$, Bonferroni-corrected Wilcoxon signed-rank).

Approach	**Strategy**	**#Refs**	**UTMOS** ↑	**Speaker SIM** ↑	**WER (%)** ↓
Single Baseline	random	1	4.425 $\pm$ 0.094	0.928 $\pm$ 0.044	15.3 $\pm$ 12.0
CONCAT	longest	1	4.390 $\pm$ 0.168	0.942 $\pm$ 0.039	22.3 $\pm$ 22.1
CONCAT	random	1	4.431 $\pm$ 0.119	0.926 $\pm$ 0.047	15.9 $\pm$ 12.7
CONCAT	longest	2	4.366 $\pm$ 0.392	0.947† $\pm$ 0.036	24.1 $\pm$ 73.6
CONCAT	random	2	4.432 $\pm$ 0.126	0.937 $\pm$ 0.042	14.4 $\pm$ 11.1
CONCAT	longest	3	4.416 $\pm$ 0.098	**0.949**† $\pm$ 0.038	21.3 $\pm$ 18.0
CONCAT	random	3	4.390 $\pm$ 0.379	0.941 $\pm$ 0.044	81.6 $\pm$ 555
CONCAT	longest	5	4.400 $\pm$ 0.143	0.948† $\pm$ 0.037	15.7 $\pm$ 13.0
CONCAT	random	5	4.385 $\pm$ 0.381	0.944† $\pm$ 0.044	39.7 $\pm$ 230
EMBED	longest	1	4.470† $\pm$ 0.086	0.936 $\pm$ 0.037	**13.4** $\pm$ 10.4
EMBED	random	1	4.462† $\pm$ 0.082	0.921 $\pm$ 0.045	14.5 $\pm$ 11.6
EMBED	longest	2	4.474† $\pm$ 0.087	0.934 $\pm$ 0.040	14.0 $\pm$ 10.8
EMBED	random	2	4.473† $\pm$ 0.072	0.927 $\pm$ 0.042	13.8 $\pm$ 11.3
EMBED	longest	3	4.469† $\pm$ 0.078	0.939 $\pm$ 0.037	15.0 $\pm$ 13.6
EMBED	random	3	**4.483**† $\pm$ 0.055	0.930 $\pm$ 0.039	14.5 $\pm$ 11.0
EMBED	longest	5	4.474† $\pm$ 0.054	0.938 $\pm$ 0.034	14.4 $\pm$ 11.9
EMBED	random	5	4.463† $\pm$ 0.103	0.937 $\pm$ 0.035	14.1 $\pm$ 11.4

Table 2: Statistically significant improvements over single baseline (Bonferroni-corrected Wilcoxon, α=0.05).

Config	n	**Metric**	p_{corr}	d
CONCAT, long.	2	SIM	.002	0.51
CONCAT, long.	3	SIM	<.001	0.60
CONCAT, long.	5	SIM	<.001	0.62
CONCAT, rand.	5	SIM	.002	0.38
EMBED, long.	1	UTMOS	<.001	0.43
EMBED, rand.	1	UTMOS	.004	0.36
EMBED, long.	2	UTMOS	<.001	0.46
EMBED, rand.	2	UTMOS	<.001	0.50
EMBED, long.	3	UTMOS	<.001	0.45
EMBED, rand.	3	UTMOS	<.001	0.61
EMBED, long.	5	UTMOS	<.001	0.55
EMBED, rand.	5	UTMOS	<.001	0.32

CONCAT, selecting longer references improves both fidelity and robustness.

For EMBED, the impact of selection strategy is smaller and inconsistent, aligning with the interpretation that embedding extraction is less sensitive to utterance duration than in-context processing.

3.5 *Per-speaker analysis*

The fidelity–naturalness trade-off is consistent across all 5 speakers, though magnitude varies. Notably, preliminary single-speaker experiments on one speaker initially suggested EMBED dominance across both metrics—a finding that did not generalise to the full 5-speaker evaluation. This highlights the importance of multi-speaker testing for drawing reliable conclusions about voice cloning strategies.

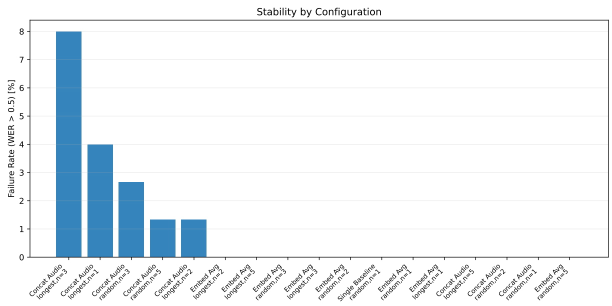

Figure 3: Stability analysis using catastrophic failure rate (WER> 50%). Failures are concentrated in CONCAT+random settings, while EMBED remains stable.

3.6 *Pilot perceptual evaluation*

A pilot blind evaluation (1 listener, 4 paired comparisons) revealed a potential metric–perception divergence. The listener consistently preferred CONCAT for voice identity but EMBED for naturalness—aligning with automated metrics on the naturalness axis. However, in a case where the SIM metric rated EMBED higher (0.983 vs. 0.976), the listener judged CONCAT as the better voice match. This suggests that cosine similarity in WavLM embedding space may not fully capture perceptual voice identity, warranting investigation with larger listener panels.

4 Discussion and Conclusion

We have presented a systematic comparison of two multi-reference voice cloning strategies for Qwen3-TTS across 1,275 controlled synthesis runs. Our central finding is a Pareto trade-off: audio concatenation optimises speaker fidelity (d up to 0.62), while embedding averaging optimises naturalness (d up to 0.61), with neither dominating both objectives.

This trade-off has practical implications. For applications prioritising speaker fidelity—audiobook narration, voice preservation—CONCAT with longest references at n=3 is preferred. For applications prioritising naturalness—virtual assistants, accessibility tools—EMBED offers more natural output with stable performance regardless of reference count or selection strategy.

Our scaling analysis reveals that reference selection strategy (longest vs. random) matters more than the number of references beyond 3. Practitioners should prioritise high-quality, longer reference utterances over accumulating additional references.

The pilot perceptual evaluation raises important questions about the validity of embedding-cosine speaker similarity as an evaluation metric. If automated SIM diverges from human perception, reported improvements in the literature may not translate to perceived quality gains. Formal listening tests with crowdsourced evaluators are needed to assess this.

Limitations. We evaluate a single TTS architecture (Qwen3-TTS 0.6B) on a single dataset (LibriTTS-R) in English only. Generalisability to other models (e.g., VALL-E [**?**], CosyVoice [**?**]), languages, and acoustic conditions remains to be established. The perceptual evaluation is preliminary (n=1) and requires validation with a statistically powered listener panel.

Future work. We plan to extend this comparison to additional TTS architectures and model sizes, conduct crowdsourced MOS and speaker similarity evaluations, explore hybrid strategies combining both approaches, and investigate cross-lingual multi-reference cloning.

Reproducibility. Code and samples will be made available upon publication.

Generative AI disclosure. A generative AI coding assistant was used for experiment scripting and manuscript preparation. All results, analysis, and scientific claims were verified by the authors.

References